

cs5800-hw_{1c}*oding_analysis*

Mingxi Zhu

May 2026

1 Coding and Analysis

1. [in Python] fib(5): Recursive Time taken: 0.0 seconds Array Time taken: 0.0 seconds Compressed Time taken: 0.0 seconds
fib(10): Recursive Time taken: 0.0 seconds Array Time taken: 0.0 seconds Compressed Time taken: 0.0 seconds
fib(15): Recursive Time taken: 0.0 seconds Array Time taken: 0.0 seconds Compressed Time taken: 0.0 seconds
fib(20): Recursive Time taken: 0.0 seconds Array Time taken: 0.0 seconds Compressed Time taken: 0.0 seconds
fib(25): Recursive Time taken: 0.01 seconds Array Time taken: 0.0 seconds Compressed Time taken: 0.0 seconds
fib(30): Recursive Time taken: 0.1 seconds Array Time taken: 0.0 seconds Compressed Time taken: 0.0 seconds
fib(35): Recursive Time taken: 0.99 seconds Array Time taken: 0.0 seconds Compressed Time taken: 0.0 seconds
[in C] fib(5): Recursive Time cost: 0.00 seconds Array Time cost: 0.00 seconds Compressed Time cost: 0.00 seconds
fib(10): Recursive Time cost: 0.00 seconds Array Time cost: 0.00 seconds Compressed Time cost: 0.00 seconds
fib(15): Recursive Time cost: 0.00 seconds Array Time cost: 0.00 seconds Compressed Time cost: 0.00 seconds
fib(20): Recursive Time cost: 0.00 seconds Array Time cost: 0.00 seconds Compressed Time cost: 0.00 seconds
fib(25): Recursive Time cost: 0.00 seconds Array Time cost: 0.00 seconds Compressed Time cost: 0.00 seconds
fib(30): Recursive Time cost: 0.00 seconds Array Time cost: 0.00 seconds Compressed Time cost: 0.00 seconds
fib(35): Recursive Time cost: 0.06 seconds Array Time cost: 0.00 seconds Compressed Time cost: 0.00 seconds
The last one under 10 sec is: fib(45): Recursive Time cost: 7.38 seconds Array Time cost: 0.00 seconds Compressed Time cost: 0.00 seconds
2. Correctness: x=33554432, y=25904487 Divide function result: 1, 7649945 Built-in function result: 1, 7649945

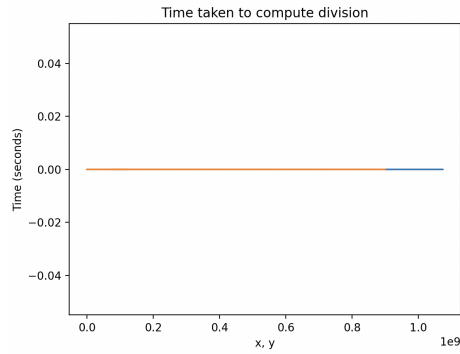


Figure 1: Enter Caption

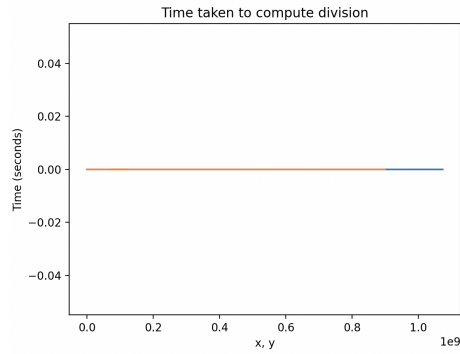


Figure 2: Enter Caption

x=67108864, y=41624518 Divide function result: 1, 25484346 Built-in function result: 1, 25484346

x=134217728, y=71080041 Divide function result: 1, 63137687 Built-in function result: 1, 63137687

x=268435456, y=104024179 Divide function result: 2, 60387098 Built-in function result: 2, 60387098

x=536870912, y=347135057 Divide function result: 1, 189735855 Built-in function result: 1, 189735855

Timing: almost constant time.